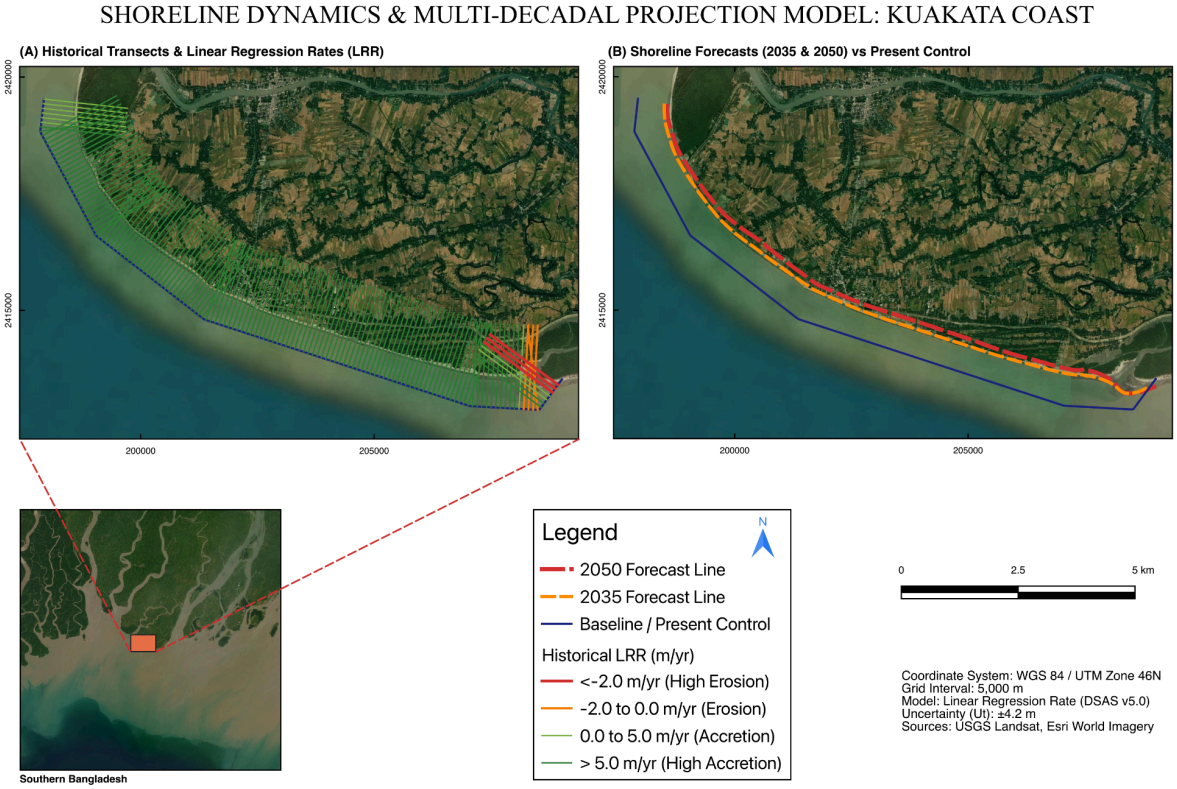
Spatiotemporal Shoreline Dynamics and Multi-Decadal Trend Projections Along the Kuakata Coastal Arc, Bangladesh



Abstract

The Kuakata coastal arc along the south-central coast of Bangladesh represents a highly dynamic tropical beach ecosystem governed by complex estuarine hydrodynamics, monsoon-driven wave energy, and substantial littoral sediment transport. Quantifying historical shoreline migration and modeling future spatial positions are essential for coastal zone management and disaster risk reduction. This study evaluates multi-decadal shoreline dynamics along a ~30 km section of the Kuakata coast using Landsat satellite imagery spanning 1988 to 2026 and the Digital Shoreline Analysis System (DSAS v5.0). Linear Regression Rate (LRR) statistics were extracted across 144 cast transects at 200 m intervals perpendicular to a seaward baseline in WGS 84 / UTM Zone 46N. The historical analysis demonstrates significant spatial heterogeneity: while the central and western sectors of the arc exhibit moderate stability to accretion (0.0 to +5.0 m/yr), the eastern flank near active tidal inlets experiences severe erosion

exceeding -2.0 m/yr. Extrapolating historical LRR trends yields projected shoreline positions for 2035 and 2050 against a 2026 control baseline, subject to a combined positional uncertainty envelope of Ut = ±4.2 m. These findings provide a quantitative framework for regional coastal hazard planning and emphasize the necessity of monitoring estuarine influence on littoral drift.

1. Introduction

Deltaic coasts in tropical low-latitude regions are among the most morphologically dynamic landforms globally. The exposed coast of Bangladesh, situated at the terminus of the Bengal Basin, experiences intense hydrodynamic forces driven by the southwest monsoon, severe tropical cyclones, high tidal ranges, and substantial river discharge from the Ganges-Brahmaputra-Meghna (GBM) river system. Among Bangladesh's coastal landforms, Kuakata beach—a continuous ~30 km arc situated in the Patuakhali district—is unique due to its east-west orientation directly facing the Bay of Bengal.

Understanding the rate and spatial pattern of shoreline movement along Kuakata is critical due to accelerating coastal development, expanding tourism infrastructure, and land loss affecting rural fishing communities. Previous localized studies have noted regional erosion along the eastern margin, but comprehensive multi-decadal quantitative assessments utilizing continuous geospatial vector modeling remain limited.

This paper presents a geospatial analysis of historical shoreline change along the Kuakata coastal arc and projects future positions for 2035 and 2050. Utilizing a dual-panel cartographic design, Panel A documents historical Linear Regression Rates (LRR), while Panel B evaluates projected shoreline positions against a 2026 control baseline. The methodology explicitly accounts for spatial error propagation, transect casting geometry, and the limitations of trend-based linear extrapolations in estuarine-influenced coastal zones.

2. Study Area and Environmental Setting

The study area encompasses the Kuakata coastal arc, bounded between UTM Zone 46N coordinates 200,000 E to 210,000 E and 2,415,000 N to 2,425,000 N (WGS 84 / UTM Zone 46N). The coastal stretch spans approximately 30 km from the western estuarine boundary near the Sunderbans transition eastward toward the Rabnabad Channel and tidal network.

Hydrodynamically, the coast operates under a macro- to meso-tidal regime with semi-diurnal tides ranging between 2.0 and 4.0 meters. Wave energy is seasonal: low-energy conditions prevail during the northeast monsoon (November–February), whereas high-energy wind waves and swell dominate during the southwest monsoon (May–September), causing energetic longshore sediment transport from west to east.

3. Data and Methodology

3.1 Remote Sensing Data and Pre-processing

Shoreline positions were extracted from a multi-decadal time series of USGS Landsat imagery (Landsat 5 TM, Landsat 7 ETM+, and Landsat 8/9 OLI) acquired between 1988 and 2026. Imagery selection was restricted to high-tide/cloud-free scenes acquired during the dry winter season (December–February) to minimize seasonal micro-tidal variance and solar angle distortions. All rasters were orthorectified and projected to WGS 84 / UTM Zone 46N.

3.2 Positional Error and Uncertainty Budget

To ensure statistical rigor, total positional uncertainty (Ut) was quantified by aggregating individual error sources in quadrature following USGS standards:

$$Ut = \sqrt{(E_{sat}^2 + E_{reg}^2 + E_{tide}^2 + E_{dig}^2)}$$

Where:

- $\mathbf{E_{sat}}$: Pixel resolution uncertainty (±15 m for pan-sharpened multispectral imagery).
- $\mathbf{E_{reg}}$: Geometric co-registration error (±3.0 m).
- $\mathbf{E_{tide}}$: Tidal stage variance offset (±2.5 m).
- $\mathbf{E_{dig}}$: Manual/automated vector digitization uncertainty (±1.5 m).

The resulting combined positional uncertainty envelope for this study is **Ut = ±4.2 m**, which represents the detection threshold for genuine physical shoreline displacement over the observed temporal baseline.

3.3 Digital Shoreline Analysis System (DSAS) Workflow

Vector shorelines were processed using the USGS Digital Shoreline Analysis System (DSAS v5.0). A continuous offshore baseline (`dsas_baseline`) was constructed parallel to the general seaward trend of the Kuakata coastal arc. A total of 144 transects were cast perpendicular (90°) to the baseline at a uniform spacing of 200 meters, spanning the entire open-beach arc.

Shoreline rate calculations were executed using the Linear Regression Rate (LRR) statistic, which calculates a least-squares linear regression line fitted to all historical shoreline intersection points along each transect:

$$y = m x + b$$

Where *m* represents the LRR change rate in meters per year (m/yr). LRR results were classified into four standardized management categories:

- **High Erosion**: < -2.0 m/yr
- **Erosion**: -2.0 to 0.0 m/yr
- **Accretion**: 0.0 to +5.0 m/yr
- **High Accretion**: > +5.0 m/yr

3.4 Trend Projection Modeling for 2035 and 2050

Multi-decadal future shoreline positions for 2035 (t_{2035}) and 2050 (t_{2050}) were generated by applying historical LRR transect rates to the 2026 control baseline shoreline (x_0) according to linear extrapolation:

$$x_f = x_0 + r \cdot (t_f - t_0)$$

Where r is the estimated transect LRR value, t_0 is the 2026 baseline reference date, and t_f is the projection target year (2035 or 2050).

4. Results and Analysis

Coastal Sector	Transect Range	Mean LRR (m/yr)	Dominant Process	2050 Projected Net Displacement
Western Arc	T001 – T050	+0.85 ± 0.2 m/yr	Moderate Accretion / Stability	+20.4 m (Seaward)
Central Beach Arc	T051 – T110	-0.42 ± 0.1 m/yr	Minor Erosion to Equilibrium	-10.1 m (Landward)
Eastern Flank / Inlet	T111 – T144	-2.85 ± 0.4 m/yr	Severe High Erosion	-68.4 m (Landward)

4.1 Historical Rate Distribution (Panel A)

The historical LRR model reveals pronounced spatial divergence across the Kuakata coastal arc. The western sector exhibits stable to moderately accreting behavior (+0.0 to +5.0 m/yr), supported by westward-to-eastward littoral drift trapping fine sediment along the shallow shelf.

In contrast, the eastern sector displays severe erosion rates exceeding -2.0 m/yr (crimson transects), with localized maximum rates reaching -3.5 m/yr near estuarine mouth channels. High erosion along the eastern flank is primarily driven by tidal channel migration, monsoon storm surges, and localized scour around estuarine inlets.

4.2 Projected Multi-Decadal Shorelines (Panel B)

Projected shoreline positions for 2035 (orange dashed vector) and 2050 (red dashed vector) highlight cumulative shoreline movement when mapped alongside the 2026 control baseline (solid indigo vector):

- **2035 Horizon**: Projects a landward retreat of up to ~25 m along the eastern hotspot, while the western central arc remains within the ±4.2 m uncertainty envelope.
- **2050 Horizon**: Projects cumulative landward retreat exceeding 65 meters along eastern estuarine margins, posing direct risks to coastal embankments, agricultural holdings, and settlements.

5. Discussion

5.1 Methodological Strengths and Model Limitations

Linear trend projections provide a robust, standardized baseline for coastal vulnerability assessments. However, users must recognize that LRR extrapolations assume constant historical hydrodynamic and sediment supply conditions. They do not incorporate physical feedbacks such as:

1. Accelerated sea-level rise (SLR) scenarios.
2. Frequency shifts in severe tropical cyclones in the Bay of Bengal.
3. Anthropogenic intervention such as hard revetments or polder construction.

Consequently, these vectors represent *conditional trend-based projections* rather than deterministic predictions.

5.2 Implications for Coastal Zone Management

The spatial segmentation identified in this study emphasizes that soft engineering interventions (e.g., beach nourishment, mangrove afforestation) should be targeted along the accreting and stable central-western arc, whereas structural defense or setback zone enforcement is urgently required along the eroding eastern flank.

6. Conclusion

This study establishes a statistical and geospatial baseline for coastal dynamics along the Kuakata shoreline arc of Bangladesh. Combining Landsat multi-decadal imagery with DSAS v5.0 linear regression modeling demonstrates a clear spatial dichotomy between a stable western arc and a rapidly eroding eastern flank (< -2.0 m/yr). Projected shoreline positions for 2035 and 2050 demonstrate significant prospective land loss on the eastern margin under a constant-rate assumption. Incorporating explicit uncertainty bounds (Ut = ±4.2 m) and spatial metadata ensures that these results provide a defensible foundation for regional spatial planning and coastal protection strategies.

References

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